ASSIGNMENT

Introduction to Asymmetric Cryptography
Cryptography and Network Security

Q1. Explain the basic principles of asymmetric (public-key) cryptography.

Meaning: Asymmetric cryptography is a method that uses a mathematically related pair of keys: a **public key** and a **private key**. The public key can be shared with everyone, while the private key is kept secret by its owner.

Roles of keys

• **Public key:** Shared openly. It can be used to encrypt information for the owner or to verify the owner's digital signature.
• **Private key:** Kept secret. It is used to decrypt information encrypted for the owner or to create a digital signature.

Security services

1. **Confidentiality:** Alice encrypts a message using Bob's public key. Only Bob's matching private key can decrypt it.
2. **Authentication:** A digital signature created with Alice's private key can be verified using Alice's public key, helping confirm the sender.
3. **Non-repudiation:** A valid digital signature provides evidence that the private-key holder signed the message. This depends on protecting the private key and using trustworthy identity procedures.

Encryption and decryption process

```
ALICE → Message + Bob's public key → Ciphertext → BOB
BOB → Ciphertext + Bob's private key → Original message
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For a digital signature, Alice signs a message digest with her private key. Bob verifies the signature with Alice's public key. In practice, secure systems use suitable signature and encryption schemes rather than simply reversing encryption.

Q2. Differentiate between symmetric and asymmetric cryptography.

Basis	Symmetric cryptography	Asymmetric cryptography
Number of keys	One shared secret key	A public/private key pair
Key distribution	Secret key must be shared securely	Public key can be shared openly; private key stays secret
Speed	Very fast; suitable for large data	Slower and more computationally expensive
Key length	Usually shorter keys for comparable practical use	Usually much longer keys; e.g. RSA 2048-bit
For n users	Needs n(n–1)/2 pairwise keys in a fully connected group	Each user generally maintains one key pair; public keys must be authenticated
Applications	File encryption, disk encryption, bulk data in TLS	Key establishment, digital signatures, certificates

Symmetric encryption is preferred for speed, while asymmetric cryptography helps solve the key-sharing and digital-signature problems. Modern systems commonly use both.

Q3. Describe the main techniques and building blocks used in public-key cryptography.

1. **Trapdoor one-way functions:** A one-way function is easy to calculate but difficult to reverse. A trapdoor is secret information that makes reversal practical for an authorized user. RSA is based on the difficulty of factoring a large composite number, together with its private-key trapdoor.

2. **Diffie–Hellman key exchange:** Two parties agree on public parameters and exchange values derived from their private choices. Each side independently calculates the same shared secret. The secret is not sent directly. Basic Diffie–Hellman does not authenticate parties by itself, so it must be combined with authentication to prevent a

man-in-the-middle attack.

3. **Digital signatures:** The sender uses a private signing key to sign a message or its hash. The receiver uses the corresponding public key to verify it. Signatures support integrity, authentication and non-repudiation when keys and identities are properly managed.

4. **Public-key infrastructure (PKI):** PKI includes certificate authorities (CAs), digital certificates, registration procedures, and certificate status/revocation systems. A certificate binds an identity to a public key and is digitally signed by a trusted CA. This helps users decide whether a public key belongs to the claimed person or server.

Q4. Trace the historical development of public-key cryptography.

• **Key distribution problem:** In symmetric cryptography, both parties need the same secret key. Sharing that key securely with many people was difficult, especially over open networks.
• **Diffie and Hellman (1976):** They published the idea of public-key cryptography and introduced the Diffie–Hellman key exchange, showing how two parties could establish a shared secret over an insecure channel.
• **RSA (1977):** Ron Rivest, Adi Shamir and Leonard Adleman developed RSA, a public-key algorithm based on the mathematical difficulty of factoring a large number into its prime factors.
• **Earlier GCHQ work:** James Ellis, Clifford Cocks and Malcolm Williamson developed related ideas inside the UK's GCHQ before the public discoveries. Their work remained classified and became publicly known later. Cocks developed an RSA-like scheme, while Williamson worked on a key-agreement method similar to Diffie–Hellman.
• **ElGamal:** Introduced in 1985, it is based on the discrete logarithm problem and supports encryption and signatures.
• **Elliptic Curve Cryptography (ECC):** Developed from elliptic-curve mathematics in the 1980s, ECC can provide comparable security to RSA with much smaller keys. It is widely used in modern protocols and devices.

Q5. Discuss advantages and disadvantages of asymmetric cryptography. Why are TLS systems hybrid?

Advantages: (1) Public keys can be shared openly, reducing the need to send secret keys in advance. (2) Digital signatures support authentication and integrity. (3) Certificates help bind identities to public keys. (4) It scales more conveniently for large groups than maintaining a separate shared secret for every pair.

Disadvantages: (1) It is slower than symmetric encryption. (2) It requires larger keys and more computing power. (3) Public keys must be authenticated to prevent impersonation. (4) Private keys must be strongly protected. (5) Poor implementations or weak parameters can make it vulnerable.

Why TLS is hybrid: A TLS connection uses asymmetric cryptography (or an authenticated key-agreement method) to authenticate the server and establish shared session keys. It then uses fast symmetric authenticated encryption, such as AES-GCM or ChaCha20-Poly1305, to protect the actual data. For example, a browser authenticates a website using its certificate, establishes session keys, and then encrypts web traffic symmetrically. This combines convenient key establishment with high-speed data protection.

Q6. Describe the RSA algorithm, key generation, encryption and decryption.

Key generation steps:
1. Choose two large distinct prime numbers, p and q.
2. Calculate the modulus: $n = p \times q$.
3. Calculate Euler's totient: $\phi(n) = (p - 1)(q - 1)$.
4. Choose public exponent e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
5. Calculate private exponent d such that $d \times e \equiv 1 \pmod{\phi(n)}$.
6. Public key is (e, n); private key is (d, n). Keep p, q and d secret.

Encryption: Convert the message into an integer M with $0 \leq M < n$. Ciphertext is $C = M^e \bmod n$.
Decryption: Recover $M = C^d \bmod n$. Real systems must use secure padding such as OAEP; raw textbook RSA should not be used directly.

Flowchart

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Choose p, q → Compute n and ϕ(n) → Choose e → Find d → Publish (e,n), protect (d,n)
      Sender: Encode + OAEP → C = M^e mod n → Receiver: M = C^d mod n → Decode
```

Why decryption works: Since $e \times d \equiv 1 \pmod{\phi(n)}$, there is an integer k such that $ed = 1 + k\phi(n)$. For valid RSA parameters and properly encoded messages, modular arithmetic (Euler's theorem and its extensions) ensures that raising the ciphertext to d recovers the original message representative. Correct padding and message encoding are essential in practical use.

Q7. What is RSA key size? Explain factoring security, common sizes, ECC comparison and quantum threat.

RSA key size is the bit length of modulus n. For example, a 2048-bit RSA key has a modulus represented using 2048 bits. RSA security relies mainly on the difficulty of factoring n into its two large prime factors p and q. If an attacker factors n, they can calculate $\phi(n)$ and derive the private key.

RSA modulus size	Approximate ECC comparable strength
1024 bits	About 160-bit ECC (both are now considered inadequate for modern protection)
2048 bits	About 224-bit ECC
3072 bits	About 256-bit ECC
4096 bits	About 300-bit ECC (approximate)

These are approximate security-strength comparisons, not identical key sizes. 1024-bit RSA is obsolete; 2048-bit RSA is a common minimum baseline, while 3072-bit RSA provides a stronger long-term margin. 4096-bit RSA is larger and slower. Follow current organizational and standards guidance for the required lifetime of the data.

Quantum threat: A sufficiently capable, fault-tolerant quantum computer running Shor's algorithm could factor RSA moduli efficiently and break RSA encryption and signatures. Large-scale machines capable of doing this are not currently available, but systems should plan migration to post-quantum cryptography.

Q8. Discuss major security considerations and attacks on RSA, padding, signatures and good practices.

1. **Factoring attacks:** If an attacker factors n into p and q, the private key can be derived. Use sufficiently large, randomly generated primes and approved key sizes.
2. **Small public exponent / small message attacks:** Using a small exponent with unsafe or repeated message encoding can expose messages. Secure randomized padding such as OAEP prevents textbook-RSA weaknesses.
3. **Common modulus attack:** Reusing the same modulus with different public exponents can create vulnerabilities under certain message-reuse conditions. Generate independent key pairs and do not share RSA moduli across users.
4. **Timing and side-channel attacks:** Execution time, power use or electromagnetic leakage may reveal information about private-key operations. Use constant-time implementations, blinding and protected hardware where appropriate.
5. **Chosen-ciphertext attacks:** An attacker may submit selected ciphertexts and learn from decryption responses. OAEP is designed to provide security against important chosen-ciphertext attacks when implemented correctly; error messages should not reveal padding details.

Padding: Raw RSA is deterministic and mathematically malleable. OAEP adds randomized encoding before encryption, preventing many structural and chosen-ciphertext attacks. For signatures, use a secure signature scheme such as RSA-PSS (or standards-compliant PKCS#1 v1.5 where required), not encryption as a substitute for signing.

RSA digital signatures: The signer hashes the message and creates a signature using the private signing key and a secure signature encoding. The receiver checks the signature using the public key and compares the verified digest with the message digest. This supports integrity and authentication; non-repudiation depends on identity proofing, key custody and applicable legal processes.

Good practices: Use at least 2048-bit RSA where RSA is required; use vetted libraries and secure random-number generation; use OAEP for encryption and PSS for signatures; protect private keys; validate certificates and key usage; avoid key/modulus reuse; keep software updated; use constant-time, side-channel-resistant implementations; and plan for post-quantum migration.

Note: These answers are written in simple assignment-style English based on the eight questions in the supplied assignment sheet. Key-size equivalences are approximate; follow your course or institution's prescribed standard if it differs.